Topic: Recursive Queries

Q. What are Recursive Queries in SQL? Explain the concept of Common Table Expressions (CTE) and how a Recursive CTE works with an example.

> **Definition to Remember**

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Recursive Queries & Common Table Expressions (CTE)

A **CTE** is a temporary, named result set defined using the `WITH` keyword. It exists only during the execution of a single SQL statement.

- Use

2. Structure of a Recursive CTE

A recursive CTE has exactly **3 components**:

```

WITH

ANCHOR MEMBER Base case (executes once, gives R)

SELEC

UNION ALL Connects anchor + recursive

RECURSIVE MEMBER References CTE itself (gives R, R, ...)

SELEC

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### 3. Working of a Recursive Query

1. **Anchor Member** executes first produces base result set R_0.

2. **Re

### 4. Example Organizational Hierarchy

**Table:** `Employee (EmpID, Name, ManagerID)` Find all levels of management.

```sql

UNION ALL

-- Recursive: Find employees managed by those already found

SELEC

## 5. Applications

- **Organizational Charts:** Employee hierarchy traversal.

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> **Must-Write Points (for 10 marks)**

> 1. R

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> **Quick Recall**

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